🎯 CSS Attribute Selectors Guide

Understanding Attribute Selectors

CSS attribute selectors allow you to target elements based on their attributes and attribute values. They enable powerful, flexible styling based on HTML attributes without adding classes or IDs. Attribute selectors are enclosed in square brackets [attribute] and can match exact values, partial values, or use pattern matching.

Key Point: Attribute selectors match elements based on HTML attributes like id, class, data-*, href, type, etc. They're case-sensitive by default but offer powerful pattern matching with ^=, $=, *=, and other operators. Combine with pseudo-classes for even more powerful selection.

Attribute Selectors Reference

Selector	Syntax	Description
[attribute]	[required]	Selects elements with the attribute (any value)
[attribute="value"]	[type="email"]	Selects elements with exact attribute value
[attribute^="value"]	[href^="https"]	Selects elements where attribute STARTS WITH value
[attribute$="value"]	[href$=".pdf"]	Selects elements where attribute ENDS WITH value
[attribute*="value"]	[href*="example"]	Selects elements where attribute CONTAINS value
[attribute~="value"]	[class~="active"]	Selects elements where attribute contains word
[attribute|="value"]	[lang|="en"]	Selects elements where attribute starts with value or value-
[attribute i]	[type="EMAIL" i]	Case-insensitive matching (i flag)

Practical Examples

Example 1: [attribute] - Presence Match

```css
/* CSS - Select all elements with required attribute */ input[required] { border: 2px solid #e74c3c; }
```

HTML Usage:

```html
<input type="text" placeholder="Optional">
<input type="text" placeholder="Required" required>
```

Live Demo:

Optional (no red border)

Required (red border)

Example 2: [attribute="value"] - Exact Match

```css
/* CSS - Select specific input types */ [type="email"] { border: 2px solid #3498db; color: #3498db; } [type="submit"] { background: #11998e; color: white; }
```

HTML Usage:

```html
<input type="text" placeholder="Text">
<input type="email" placeholder="Email">
<input type="submit" value="Submit">
```

Live Demo:

Text input

Email input

Submit button

Example 3: [attribute^="value"] - Starts With

```css
/* CSS - Style HTTPS links differently */ a[href^="https"] { color: #11998e; font-weight: bold; position: relative; } a[href^="https"]::before { content: "🔒 "; }
```

HTML Usage:

```html
<a href="https://secure.example.com">Secure Link</a>
<a href="http://example.com">Regular Link</a>
```

Live Demo:

🔒 **Secure HTTPS Link**

Regular HTTP Link

Example 4: [attribute$="value"] - Ends With

```css
/* CSS - Style PDF links differently */ a[href$=".pdf"] { color: #e74c3c; font-weight: bold; } a[href$=".pdf"]::after { content: " [PDF]"; font-size: 12px; }
```

HTML Usage:

```html
<a href="document.pdf">Download PDF</a>
<a href="page.html">Web Page</a>
```

Live Demo:

Download PDF [PDF]

Web Page

Example 5: [attribute*="value"] - Contains

```css
/* CSS - Highlight links containing "example" */ a[href*="example"] { background: #ecf0f1; padding: 2px 4px; border-radius: 3px; }
```

HTML Usage:

```html
<a href="https://example.com">Example.com</a>
<a href="https://my-example.org">My-Example.org</a>
<a href="https://google.com">Google</a>
```

Live Demo:

Example.com

My-Example.org

Google

Example 6: [data-*] - Custom Data Attributes

```css
/* CSS - Style based on data attributes */ [data-status="active"] { background: #27ae60; color: white; padding: 10px; border-radius: 4px; } [data-status="inactive"] { background: #e74c3c; color: white; padding: 10px; border-radius: 4px; }
```

HTML Usage:

```html
<div data-status="active">Active</div>
<div data-status="inactive">Inactive</div>
```

Live Demo:

Active Status

Inactive Status

Example 7: [data-*] - Priority Levels

```css
/* CSS - Color-coded priorities */ [data-priority="high"] { border-left: 4px solid #e74c3c; background: #ffebee; } [data-priority="medium"] { border-left: 4px solid #f39c12; background: #fff8e1; } [data-priority="low"] { border-left: 4px solid #27ae60; background: #e8f5e9; }
```

HTML Usage:

```html
<div data-priority="high">High Priority Task</div>
<div data-priority="medium">Medium Priority</div>
<div data-priority="low">Low Priority</div>
```

Live Demo:

High Priority Task

Medium Priority Task

Low Priority Task

Example 8: [attribute~="value"] - Word Match

```css
/* CSS - Match class as word */ [class~="featured"] { border: 3px solid gold; box-shadow: 0 0 10px rgba(255, 215, 0, 0.5); }
```

HTML Usage:

```html
<div class="featured">Featured Item</div>
<div class="featured-item">Not featured (won't match)</div>
```

Live Demo:

Featured Item

Featured-item (not matched)

Example 9: [attribute|="value"] - Language Match

```css
/* CSS - Match language codes */ [lang|="en"] { color: #2c3e50; } [lang="en-US"] { font-weight: bold; }
```

HTML Usage:

```html
<p lang="en">English</p>
<p lang="en-US">US English</p>
<p lang="en-GB">British English</p>
```

Live Demo:

English
US English (bold)
British English

Example 10: [attribute i] - Case-Insensitive

```css
/* CSS - Case-insensitive matching */ input[type="text" i] { border: 2px solid #11998e; } /* Matches: type="text", type="TEXT", type="Text" */
```

HTML Usage:

```html
<input type="text">
<input type="TEXT">
<input type="Text">
```

Live Demo:

All inputs styled with [type="text" i]:

type=text

Example 11: Combining Attribute Selectors

```css
/* CSS - Multiple conditions */ input[type="text"][required] { border: 3px solid #e74c3c; box-shadow: inset 0 0 5px rgba(231, 76, 60, 0.2); } a[href^="https"][href*="secure"] { color: #27ae60; font-weight: bold; }
```

HTML Usage:

```html
<input type="text" required placeholder="Required text">
<input type="text" placeholder="Optional text">
<a href="https://secure.example.com">Secure Link</a>
```

Live Demo:

Required text input

Optional text input

Secure HTTPS Link

Attribute Selector Patterns

Pattern	Syntax	Matches	Example
Exact match	[attr="value"]	Exact value only	[type="email"]
Starts with	[attr^="value"]	value at beginning	[href^="https"]
Ends with	[attr$="value"]	value at end	[src$=".png"]
Contains	[attr*="value"]	value anywhere	[class*="btn"]
Word match	[attr~="value"]	value as whole word	[class~="active"]
Language match	[attr|="value"]	value or value-	[lang|="en"]

Common Attribute Selector Use Cases

Use Case	Selector	Purpose
External links	[href^="http"]	Style links to external sites
Secure links	[href^="https"]	Highlight secure/safe links
File downloads	[href$=".pdf"]	Indicate downloadable files
Button types	[type="submit"]	Style specific form buttons
Data attributes	[data-status]	Style elements by custom data
Required fields	input[required]	Highlight form validation
Disabled state	input[disabled]	Style inactive form elements
Language	[lang|="en"]	Style by language code

Best Practices

1. Use Data Attributes Effectively: Leverage [data-*] attributes for styling hooks. This keeps HTML semantic and CSS maintainable.

2. Combine with Other Selectors: Mix attribute selectors with pseudo-classes (input[type="text"]:focus) for powerful, specific targeting.

3. Remember Case Sensitivity: Attribute selectors are case-sensitive by default. Use [attr="value" i] for case-insensitive matching when needed.

4. Avoid Over-specificity: Attribute selectors have medium specificity (0,1,0). Don't use too many combined selectors that become hard to override.

5. Document Custom Attributes: When using [data-*] attributes, document them clearly. They're not standard HTML and need clear intent.

Common Issues & Solutions

Issue 1: Case Sensitivity Breaking Matching
Problem: [type="TEXT"] doesn't match type="text"
Solution: Use [type="text" i] flag for case-insensitive matching.

Issue 2: Selector Not Working with Dashes
Problem: [data-my-attr="value"] seems correct but doesn't match
Solution: Dashes are allowed in attribute names. Check attribute spelling and value exactly.

Issue 3: Word Match (~=) Not Working
Problem: [class~="active"] doesn't match class="is-active"
Solution: ~= matches whole words only. "active" and "is-active" are different words. Use *="active" instead.

Specificity of Attribute Selectors

Selector Type	Specificity	Example
Attribute selector	0, 1, 0	[type="text"]
Class selector	0, 1, 0	.button
Element + attribute	0, 1, 1	input[type="text"]
Multiple attributes	0, 2, 0	[type="text"][required]
ID selector	1, 0, 0	#myElement

Summary

CSS attribute selectors provide powerful, semantic ways to target and style elements based on their attributes without adding extra classes or IDs. Understanding the different matching patterns (exact, starts with, ends with, contains, word match) enables you to create flexible, maintainable stylesheets. Combined with data attributes and pseudo-classes, attribute selectors allow for sophisticated styling logic directly in CSS.

Key Takeaway: Attribute selectors match elements based on HTML attributes with optional pattern matching. Use [data-*] attributes for semantic styling hooks. Remember they're case-sensitive by default (use 'i' flag to change). Combine with pseudo-classes for powerful, specific selectors. Attribute selectors have the same specificity as class selectors.